



Course: Probability and Statistics

Time Allowed: 3 hours

Maximum marks: 50

Q.1: Suppose that the 4 inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John who stamps 20% of the packages, fails to stamp the expiration date twice in every 100 packages; Tom who stamps 60% of the packages, fails to stamp once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 150 packages. If a customer complains that his package of film does not show the expiration date, what is the probability that it was inspected by Tom? (5)

Q.2: Consider the density function: (8)

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Evaluate k (b) Find Mean and Variance of X .

(c) Find $F(x)$ and use it to evaluate $P(0.2 < x < 0.6)$.

Q.3: Suppose that X and Y are independent random variables having the joint probability distribution: (6)

$f(x,y)$		x	
		2	4
y	1	0.10	0.15
	3	0.20	0.30
	5	0.10	0.15

Find (a) $E(2X - 3Y)$; (b) $E(XY)$.

Q.4: A manufacturing company uses an acceptance scheme on items from a production line before they are shipped. The plan is a two-stage one. Boxes of 25 items are readied for shipment, and a sample of 3 items is tested for defectives. If any defectives are found, the entire box is sent back for 100% screening. If no defectives are found, the box is shipped.

(a) What is the probability that a box containing 3 defectives will be shipped?

(b) What is the probability that a box containing only 1 defective will be sent back for screening? (5)

Q.5: Suppose the probability that any given person will believe a tale about the transgressions of a famous actress is 0.7. What is the probability that:

(a) The fifth person to hear this tale is the third one to believe it.

(b) The fourth person to hear this tale is the first one to believe it. (4)

P.T.O.

FORMULAS of STATISTICS

* = multiplication

Rule of addition of probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Rule of multiplication of probability: $P(A \cap B) = P(A)P(B|A)$

Rule of subtraction of probability: $P(A^c) = 1 - P(A)$

Random Variables: X and Y are random variables, a and b are constants.

Expected value of $X = E(X) = \mu_x = \sum (x_i * P(x_i))$, Variance of $X = Var(X) = \sigma^2 = \sum (x_i - E(x))^2 * P(x_i) = \sum (x_i - \mu_x)^2 * P(x_i)$

Normal random variable = $z = (x - \mu) / \sigma$

Chi-square statistic = $\chi^2 = [(n-1)S^2] / \sigma^2$, t statistic = $(\bar{x}_1^2 \sigma_1^2) / (\bar{x}_2^2 \sigma_2^2)$

$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$

Expected value of sum of random variables = $E(X + Y) = E(X) + E(Y)$

Expected value of difference between random variables = $E(X - Y) = E(X) - E(Y)$

Variance of the sum of independent random variables = $Var(X + Y) = Var(X) + Var(Y)$

Variance of difference between independent random variables = $Var(X - Y) = Var(X) + Var(Y)$

Binomial formula: $P(X = x) = b(x; n, p) = {}^nC_x * p^x * (1-p)^{n-x} = {}^nC_x * p^x * (q)^{n-x}$

Mean of binomial distribution = $\mu_x = n * p$, Variance of binomial distribution = $\sigma_x^2 = np(1-p)$

Negative binomial formula: $P(X = x) = b^*(x; r, p) = {}^{x-1}C_{r-1} * p^r * (1-p)^{x-r}$

Mean of negative binomial distribution = $\mu_x = r/q/p$, Variance of negative binomial distribution = $\sigma_x^2 = r/q/p^2$

Geometric formula: $P(X = x) = g(x; p) = p * q^{x-1}$

Mean of geometric distribution = $\mu_x = q/p$, Variance of geometric distribution = $\sigma_x^2 = q/p^2$

Hypergeometric formula: $P(X = x) = h(x; N, n, k) = ({}^kC_x)({}^{N-k}C_{n-x}) / ({}^NC_n)$

Mean of hypergeometric distribution = $\mu_x = [n * k] / N$

Variance of hypergeometric distribution = $\sigma_x^2 = n * k * (N-k) * (N-n) / [N^2 * (N-1)]$

Poisson formula: $P(x; \mu) = [e^{-\mu} \mu^x] / x!$

Mean of Poisson distribution = $\mu_x = \mu$, Variance of Poisson distribution = $\sigma_x^2 = \mu$

Multinomial formula: $P = [n! / n_1! n_2! \dots n_k!] * (p_1^{n_1} * p_2^{n_2} * \dots * p_k^{n_k})$

Hypothesis Testing:

Standardized test statistic = (Statistic - Parameter) / (Standard deviation of statistic)

One sample z-test for proportions: z-score = $z = (p - p_0) / \sqrt{p * q / n}$

Two sample z-test for proportions: z-score = $z = (\hat{p}_1 - \hat{p}_2) / \sqrt{\hat{p} \hat{q} (\frac{1}{n_1} + \frac{1}{n_2})}$

One sample t-test for means: t-score = $t = (\bar{x} - \mu) / (s / \sqrt{n})$

Two-sample t-test for means, unknown but equal variances: t-score = $t = [(\bar{x}_1 - \bar{x}_2) - d_0] / S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

Confidence interval for μ , σ known: $\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

Confidence interval for $\mu_1 - \mu_2$, $\sigma_1^2 = \sigma_2^2$ but unknown, $n_1, n_2 \leq 30$, is $(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}} (v) S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

Course: Probability and Statistics (for CS)

Time Allowed: 3 hours

Maximum marks: 50

Note: Attempt all questions.

Q1: Write short answers to the following questions. (8)

- Differentiate between Sample and Population.
- Define Statistical Inference.
- Write any two properties of Normal Distribution.
- What is the difference between one-tailed test and two-tailed test?

Q2: In a high school graduating class of 110 students, 58 studied Mathematics, 72 studied Statistics and 40 studied both Statistics and Mathematics. If one of these students is selected at random, find the probability that:

- the student took Mathematics or Statistics;
- the student took Statistics but not Mathematics.

Q3: Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as:

f(x,y)	x			
	1	2	3	
y	1	0.05	0.05	0.1
	2	0.05	0.1	0.35
	3	0	0.2	0.1

- Find $P(Y=3/X=2)$,
- Find the covariance of the random variables X and Y ,
- Show whether X and Y are independent or not.

Q4: Suppose the probability is 0.7 that any given person will believe a tale about the transgressions of a famous actress. What is the probability that:

- the fifth person to hear this tale is the third one to believe it?
- the fourth person to hear this tale is the first one to believe it?

Q.6: The average life of a certain type of small motor is 10 years with a S.D. of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If he is willing to replace only 4% of the motors that fail, how long a guarantee should be offered? Assume that the lifetime of a motor follows a normal distribution. (5)

Q.7: The random variable X , representing the number of cherries in a cherry puff, has the following probability distribution:

x	4	5	6	7
$P(X=x)$	0.2	0.4	0.3	0.1

(a) Find the mean μ and the variance σ^2 .

(b) Find the mean $\mu_{\bar{x}}$ and the variance $\sigma_{\bar{x}}^2$ of the mean $\bar{x}$ for random samples of 36 cherry puffs.

(c) Find the probability that the average number of cherries in 36 cherry puffs will be less than 5.1 or greater than 5.6. (6)

Q.8: A random sample of 12 shearing pins is taken in a study of the Rockwell hardness of the pin head. Measurements on the Rockwell hardness are made for each of the 12, yielding an average value of 48.50 with a sample S.D. of 1.5. Assuming the measurements to be normally distributed:

(a) Construct 95% confidence interval for the mean Rockwell hardness.

(b) What will be the 95% confidence interval if population variance is 4? (6)

Q.9: A manufacturer claims that the average tensile strength of thread A exceeds the tensile strength of thread B by at least 10 kg. To test this claim, 60 pieces of each type of thread were tested under similar conditions. Type A thread had an average tensile strength of 87 kg with a S.D. of 6 kg, while thread B had an average tensile strength of 78 with a S.D. of 5 kg. Test the manufacturer's claim using a 0.05 level of significance. (5)

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Probability & Random Variable	Course Code:	EE 319
Program:	BS (Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	60
Paper Date:	7-Apr-17	Weight	15%
Section:	ALL	Page(s):	1
Exam:	Sessional 2	Questions	3

Instruction/Notes:

- Show complete working.
- Proceed systematically.

Q1 Determine the value of M, mean, variance and the CDF for the following probability distribution function. [15] CLO 03

$$f(x) = \begin{cases} \frac{x^2}{M} & , 0 < x < 3 \\ 0 & , otherwise \end{cases}$$

Q2 During transmission one bit in 1000 gets corrupted, causing bit error. What is the probability that at the most 2 bit are corrupted in the transmission of a block of 3000 bits. [15] CLO 04

Q3 Thirty cars pass through the gate in 15 minutes. What is the probability that the next car will pass after 1 minute. [15] CLO 05

Handwritten notes: $t = 15 \text{ min}$, $\lambda = 2?$, $1 - e^{-4}$

Q4 A manufacturer produces 1K Ω resistors. The resistors are acceptable if their value is within 10% of the rated 1K Ω . What fraction of resistances is rejected? Value of resistance is normally distributed with a standard deviation of 50 Ω . [15] CLO 05

Handwritten note: 0.036 ?

Handwritten note: $P(900 < X < 1.1 \text{ k}) = \text{success}$

Handwritten note: $\frac{X - \mu}{\sigma} = Z$

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Probability & Random Variable
Program:	BS (Electrical Engineering)
Duration:	60 Minutes
Paper Date:	21-Feb-17
Section:	ALL
Exam:	Sessional 1

Course Code:	EE 319
Semester:	Spring 2017
Total Marks:	60
Weight	15%
Page(s):	1
Questions	3

Instruction/Notes:

- Define and name all events clearly.
- Show complete working.

L15 - 4522
EE - 2A

- Q1** If sampling space is $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and all sampling points are equally likely to occur, find the $P(B|A)$ where event $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 3, 4, 5\}$ [15] CLO 01
- Q2** A box contains 5 ceramic thermistors and 7 ceramic capacitors. If we pick two pieces randomly what is the probability that at least one of them is a capacitor. [15] CLO 01
- Q3** Resistances are purchased from three companies A, B, and C. A supplies twice as many resistances as B. B and C supply same number of resistances. Historical data shows that the percentage of defective items in the supplies of A, B and C are respectively 2%, 1% and 3%. All these resistances are placed into one box. If one item is randomly chosen from the box, what is the probability that it is defective? [15] CLO 02
- Q4** A study shows that 54% of 18 year olds in Pakistan are boys and 25% of boys and 15% of girls suffer from asthma. An 18 year youth is observed with asthma, what is the probability that she is a girl. [15] CLO 02

$$P\left(\frac{C}{B}\right)$$

OR — U
And — n

(Midterm II) Fall 2015 MT-206 Probability and Statistics (CS) Total marks: 45

Date: 3rd November, 2015 Time: 90 min.

Note: Exchange of calculators and stationary is strictly prohibited. Attempt all questions. Show each step clearly.

Question 1: (10 marks) Suppose that a person's score X on a mathematics aptitude test is a number between 0 and 1, and that his score Y on a music aptitude test is also a number between 0 and 1. Suppose further that in the population of all college students in the United States, the scores X and Y are distributed according to the following joint p.d.f.:

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

- If a student's score on the music test is 0.3, what is the probability that his score on the mathematics test will be greater than 0.8?
- Are X and Y independent variables? Explain.

Question 2: (5 marks) Let the random variable X denote the number of network blackouts in a day. The probability mass function of X is:

X	0	1	2
$P(X)$	0.7	0.2	0.1

A small internet trading company estimates that each network blackout results in a \$500 loss. Find the expected loss and variance of the loss incurred in a day due to blackouts.

Question 3: (5 marks) During a particular period a university's information technology office received 20 service orders for problems with printers, of which 8 were laser printers and 12 were inkjet models. A sample of 5 of these service orders is to be selected for inclusion in a customer satisfaction survey. What is the probability that at least 3 of the selected service orders were for inkjet printers?

Question 4: (10 marks) A new computer virus attacks a folder. Each file gets damaged with probability 0.2 independently of other files.

- What is the probability that fewer than 11 files get damaged if the folder consisted of 15 files?
- What is the probability that the fourth file attacked will be the third one to get damaged?

Question 5: (5 marks) The number of computer shutdowns during any month averaging 0.25 shutdowns per month. What is the probability of at least 3 computer shutdowns during the next year?

Question 6: (10 marks) Upgrading a certain software package requires installation of new files. Files are installed consecutively. The installation time of each file is random and is normally distributed, but on the average, it takes 15 sec to install one file, with a standard deviation of 3.317 seconds.

- What is the probability that a file takes more than 5 seconds to install?
- If a file is in the top 10% of the files which take most installation time, how much time the file must have taken?